KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

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Not for students

MATHEMATICS

Year 9 5.2

Yearly Exam

2026

Marking Guide

Notes for markers

1. Full marks for a correct answer unless the question asks for working.
2. Follow-through: a later part that uses an earlier wrong answer correctly earns its marks.
3. Units: do not deduct for missing units; note it for feedback.
4. Rounding: deduct only where the criteria say so.
5. Green = Foundation, Blue = Standard (Test yourself levels).

Sections and Marks

Section 1 - Data
5 green · 2 blue /7

Section 2 - Surface area and volume
5 green · 2 blue /7

Section 3 - Mixed: Chapters 1-8
4 green · 2 blue /6

Total marks /20

Section	Marks	Share	Green	Blue	Questions
Section 1 - Data	7	35%	5	2	1 MC · 1 SA · 1 WP
Section 2 - Surface area and volume	7	35%	5	2	1 MC · 1 SA · 1 WP
Section 3 - Mixed: Chapters 1-8	6	30%	4	2	1 MC · 1 SA · 1 WP
Total	**20**	**100%**	**14 (70%)**	**6 (30%)**	

Section 1: Data: answer summary

Q	Answer	Marks	Level	Topic and source
1	B (4)	1	Green	Mode from a dot plot PLACEHOLDER (TY9 Q1 style)
2a	38	1	Green	Stem-and-leaf plot: median and range PLACEHOLDER (TY9 Q3 style)
2b	33	1	Green	Stem-and-leaf plot: median and range PLACEHOLDER (TY9 Q3 style)
3a	20	1	Green	Frequency table and histogram PLACEHOLDER (TY9 Q4 style)
3b	1 pet	1	Green	Frequency table and histogram PLACEHOLDER (TY9 Q4 style)
3c	1.5 pets	2	Blue	Frequency table and histogram PLACEHOLDER (TY9 Q4 style)
	Total	**7**	5 green · 2 blue	

Section 2: Surface area and volume: answer summary

Q	Answer	Marks	Level	Topic and source
1	C (125 cm^3)	1	Green	Volume of a cube PLACEHOLDER (TY10 style)
2	552.9 cm^3	2	Green	Volume of a cylinder PLACEHOLDER (TY10 style)
3a	0.48 m^3	1	Green	Volume, capacity and surface area of a prism PLACEHOLDER (TY10 style)
3b	480 L	1	Green	Volume, capacity and surface area of a prism PLACEHOLDER (TY10 style)
3c	2.96 m^2; \$133.20	2	Blue	Volume, capacity and surface area of a prism PLACEHOLDER (TY10 style)
	Total	**7**	5 green · 2 blue	

Section 3: Mixed: Chapters 1-8: answer summary

Q	Answer	Marks	Level	Topic and source
1	B (a^8)	1	Green	Index laws PLACEHOLDER (TY5 style)
2	13.0 cm	2	Green	Pythagoras' theorem: hypotenuse PLACEHOLDER (TY2 style)
3a	\$934.80	1	Green	Wages and overtime PLACEHOLDER (TY8 style)

Section 3: Mixed: Chapters 1-8: answer summary (continued)				
Q	**Answer**	**Marks**	**Level**	**Topic and source**
3b	$1082.40	2	Blue	Wages and overtime PLACEHOLDER (TY8 style)
	Total	**6**	4 green · 2 blue	

Section 1: Data: marking criteria

Question 1 1 mark · Green · Mode from a dot plot — PLACEHOLDER (TY9 Q1 style)

ANSWER **B (4)**
MARKS **1 mark:** B (4)
NOTES *4 occurs most often (4 times).*

Question 2 2 marks · Green · Stem-and-leaf plot: median and range — PLACEHOLDER (TY9 Q3 style)

a) ANSWER **38** — 1 mark · Green
WORKING 11 values: the median is the 6th value
MARKS **1 mark:** median 38

b) ANSWER **33** — 1 mark · Green
WORKING 56 − 23
MARKS **1 mark:** range 33

Question 3 4 marks · Green + Blue · Frequency table and histogram — PLACEHOLDER (TY9 Q4 style)

a) ANSWER **20** — 1 mark · Green
WORKING 4 + 7 + 6 + 2 + 1
MARKS **1 mark:** 20

b) ANSWER **1 pet** — 1 mark · Green
MARKS **1 mark:** 1 pet (the highest column)

c) ANSWER **1.5 pets** — 2 marks · Blue
WORKING Sum of $f \times x = 0 + 7 + 12 + 6 + 4 = 29$
Mean = $\frac{29}{20}$
MARKS **1 mark:** correct sum of f × x (follow-through from the table)
1 mark: 1.5
NOTES *Accept 1.45. Follow-through from part a.*

Section 2: Surface area and volume: marking criteria

Question 1 1 mark · Green · Volume of a cube — PLACEHOLDER (TY10 style)

ANSWER **C (125 cm^3)**
MARKS **1 mark:** C (125 cm^3)

Question 2 2 marks · Green · Volume of a cylinder — PLACEHOLDER (TY10 style)

ANSWER **552.9 cm^3**
WORKING $V = \pi \times 4^2 \times 11$
MARKS **1 mark:** correct substitution into $V = \pi r^2 h$
1 mark: 552.9 cm^3

Marking criteria (continued)

Question 3 4 marks ■ Green + ● Blue Volume, capacity and surface area of a prism PLACEHOLDER (TY10 style)

a) 1 mark · ■ Green

ANSWER **0.48 m³**

WORKING 1.2 × 0.8 × 0.5

MARKS **1 mark:** 0.48 m³

b) 1 mark · ■ Green

ANSWER **480 L**

WORKING 0.48 × 1000

MARKS **1 mark:** 480 L (follow-through from a)

c) 2 marks · ● Blue

ANSWER **2.96 m²; $133.20**

WORKING Base 1.2 × 0.8 = 0.96

Sides 2 × 1.2 × 0.5 + 2 × 0.8 × 0.5 = 2

MARKS **1 mark:** area of glass 2.96 m²

1 mark: cost $133.20 (follow-through)

Section 3: Mixed: Chapters 1-8: marking criteria

Question 1 1 mark ■ Green Index laws PLACEHOLDER (TY5 style)

ANSWER **B (a^8)**

MARKS **1 mark:** B (a^8)

Question 2 2 marks ■ Green Pythagoras' theorem: hypotenuse PLACEHOLDER (TY2 style)

ANSWER **13.0 cm**

WORKING $x^2 = 7^2 + 11^2 = 170$

$x = \sqrt{170}$

MARKS **1 mark:** $x^2 = 170$

1 mark: 13.0 cm

Question 3 3 marks ■ Green + ● Blue Wages and overtime PLACEHOLDER (TY8 style)

a) 1 mark · ■ Green

ANSWER **$934.80**

WORKING 38 × $24.60

MARKS **1 mark:** $934.80

b) 2 marks · ● Blue

ANSWER **$1082.40**

WORKING Overtime: 4 × 1.5 × $24.60 = $147.60

$934.80 + $147.60

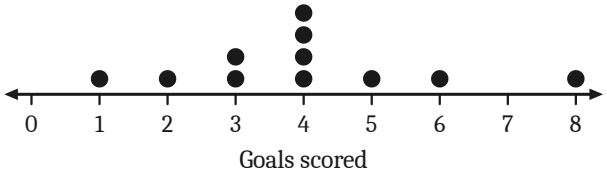
MARKS **1 mark:** overtime pay $147.60

1 mark: total $1082.40 (follow-through from a)

Marking guide: answers in red

Section 1: Data

Multiple Choice:

		Mark
1	[PLACEHOLDER] The dot plot shows the number of goals scored by a netball team in each game. What is the mode? A. 7 **B. 4** C. 5 D. 11 	1

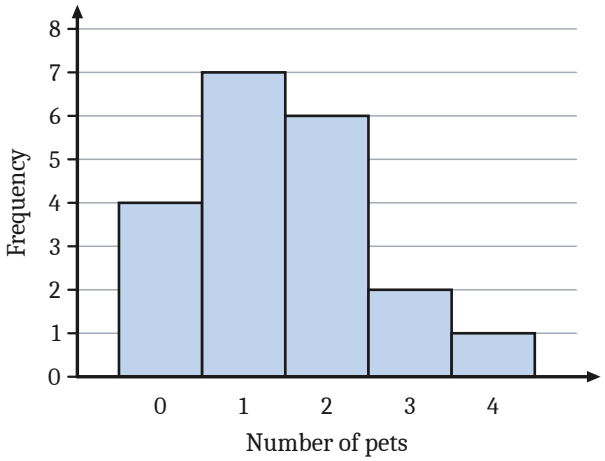
Short Answer:

	Question	Working Out	Mark
2	[PLACEHOLDER] The stem-and-leaf plot shows the ages of the people at a family lunch. Stem | Leaf 2 | 3 5 3 | 1 4 4 8 4 | 2 5 7 5 | 1 6 Key: 2 | 3 = 23 a. Find the median age. b. Find the range.	a. 11 values: the median is the 6th value **38** b. 56 – 23 **33**	1 1

Marking guide: answers in red

Section 1: Data (continued)

Worded Problems:

	Question	Working Out	Mark
3	[PLACEHOLDER] A group of Year 9 students were asked how many pets they have. The results are shown in the frequency table and the histogram. **Pets (x) / Frequency (f)** 0 — 4 1 — 7 2 — 6 3 — 2 4 — 1 		
a	How many students were surveyed?	4 + 7 + 6 + 2 + 1 **20**	1
b	What is the mode?	**1 pet**	1
c	Find the mean number of pets, correct to 1 decimal place.	Sum of $f \times x = 0 + 7 + 12 + 6 + 4 = 29$ Mean = $\frac{29}{20}$ **1.5 pets**	2
	Total		**/7**

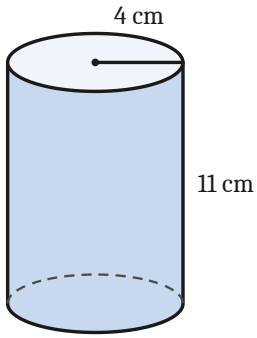
Marking guide: answers in red

Section 2: Surface area and volume

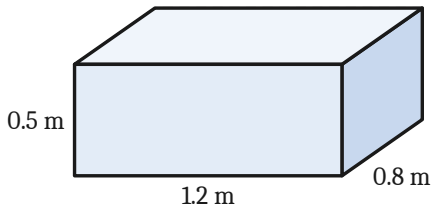
Multiple Choice:

		Mark
1	[PLACEHOLDER] What is the volume of a cube with sides of 5 cm? A. 15 cm^3 B. 25 cm^3 **C. 125 cm^3** D. 150 cm^3	1

Short Answer:

	Question	Working Out	Mark
2	[PLACEHOLDER] Find the volume of this cylinder, correct to 1 decimal place.  Note: Volume of a cylinder: $V = \pi r^2 h$	$V = \pi \times 4^2 \times 11$ **552.9 cm^3**	2

Worded Problems:

	Question	Working Out	Mark
3	[PLACEHOLDER] A fish tank is a rectangular prism with an open top.  		
a	Find the volume of the tank in m^3.	$1.2 \times 0.8 \times 0.5$ **0.48 m^3**	1
b	How many litres of water does the tank hold? (1 m^3 = 1000 L)	0.48×1000 **480 L**	1

Marking guide: answers in red

	Section 2: Surface area and volume (continued)		
	Question	**Working Out**	**Mark**
c	Glass for the tank costs \$45 per m^2. Find the area of glass needed (no lid) and its cost.	Base $1.2 \times 0.8 = 0.96$ Sides $2 \times 1.2 \times 0.5 + 2 \times 0.8 \times 0.5 = 2$ **2.96 m^2; \$133.20**	2
	Total		**/7**

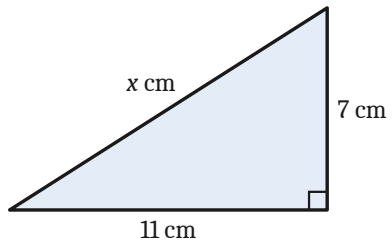
Marking guide: answers in red

Section 3: Mixed: Chapters 1-8

Multiple Choice:

		Mark
1	[PLACEHOLDER] Simplify $a^5 \times a^3$. A. a^2 **B. a^8** C. a^{15} D. $2a^8$	1

Short Answer:

	Question	Working Out	Mark
2	[PLACEHOLDER] Find the value of x, correct to 1 decimal place.  Note: Pythagoras' theorem: $c^2 = a^2 + b^2$	$x^2 = 7^2 + 11^2 = 170$ $x = \sqrt{170}$ **13.0 cm**	2

Worded Problems:

	Question	Working Out	Mark
3	[PLACEHOLDER] Mia works at a garden centre for \$24.60 per hour. Last week she worked 38 hours at the normal rate and 4 hours of overtime at time-and-a-half.		
a	Find her pay for the normal hours.	$38 \times \$24.60$ **\$934.80**	1
b	Find her total pay for the week.	Overtime: $4 \times 1.5 \times \$24.60 = \$147.60$ $\$934.80 + \$147.60$ **\$1082.40**	2
	Total		**/6**